

Khushi Kalpesh Joshi

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SUMMARY

CS (Data Science) undergraduate shipping across ML and full-stack: built a GNN achieving 0.033 eV/atom MAE on crystal property prediction and production-style REST APIs powering OCR data pipelines. Founder & CEO of a pan-India social-impact community; has led 50+ person volunteer teams and organized 7+ tech and impact events reaching 120+ participants.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

Machine Learning: PyTorch, PyTorch Geometric, Scikit-learn, Pandas, NumPy, Matplotlib

Backend / Web: FastAPI, Flask, REST APIs (OpenAPI), React, HTML, CSS, JavaScript, Bootstrap

Tools: Git, Jupyter Notebook

PROJECTS

GNN for Crystal Property Prediction | *PyTorch, PyTorch Geometric, Materials Project API*

- Developed a multi-scale GNN using 3 cutoff radii (4Å, 6Å, 8Å) with cross-attention fusion, achieving MAE 0.033 eV/atom and R^2 0.996 on held-out test sets
- Resolved WL isomorphism collisions across 10 material pairs in TiO₂, SnO₂, and Fe₂O₃ systems where distinct crystal structures appear identical to standard GNNs

CarbonDrop | *FastAPI, React, Tailwind, SQLite, Tesseract OCR, RapidFuzz* | GitHub

2025 – Present

- Built a multi-domain carbon footprint calculator processing receipts, bills, and tickets via OCR, fuzzy-matching items against DEFRA/EPA emission datasets for per-item CO₂ figures
- Engineered 6 what-if simulation endpoints, an EcoCredits gamification layer, and a Chart.js dashboard; documented the REST API via auto-generated OpenAPI spec

Email Phishing Detection | *Python, Scikit-learn*

- Built a supervised ML classifier to flag phishing emails, engineering text features and running cross-validated evaluation to select the best model

Movie Recommendation System | *Python, Scikit-learn, Streamlit*

- Implemented a TF-IDF based content recommendation engine with an interactive Streamlit UI for exploring similar titles

EDUCATION

AMC Engineering College, Bengaluru

2023 – 2027

B.E. Computer Science (Data Science)

LEADERSHIP & EXPERIENCE

Founder & CEO, COMPASSION CREW

2026 – Present

Social-impact community: education, women's empowerment, animal welfare, community development

- Founded and lead a pan-India community connecting students, professionals, and changemakers through expert talks, events, and volunteering
- Co-organized "Intelligent Healthcare: The Rise of Agentic AI" with Moksha Foundation at PES University, coordinating an industry AI speaker and full-day agenda

Organizing Team, SheBuilds Bangalore

2025 – 2026

Media & Sponsorship (2025); Event Lead (2026)

- Handled sponsorship outreach (2025); organized 3 expert sessions in 2026 – Founders Talk, MCP & AI Agents Workshop, TEDx-style Tech Talk – managing speakers and logistics
- Supported an AI & Vibe Coding Workshop with Compassion Crew at PES College, Banashankari

College Technical Events

2026

Designer / Organizer

- Designed a 3-level browser game for Tech-O-Ween (25 teams) and ran a campus esports tournament with 120+ participants

Volunteer Coordinator, Lok Sabha Elections

2024

Managed 50+ volunteers and polling operations

ACHIEVEMENTS

Ontario Math Olympiad — 1st Place • Robotics Competition Finalist • INTSO Merit Medals (Math, Science, English)

ADDITIONAL

NCC: Leading Flight Cadet (B & C Certificate) | **Languages:** English, Hindi, Gujarati, French (B2)